

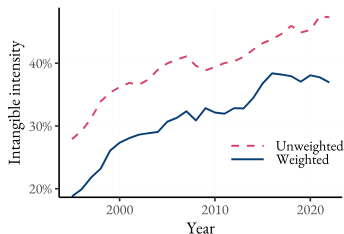
A Quantitative Theory of the Rise of Intangibles, Multinational Production, and Profit Shifting

S. Dyrda¹, G. Hong², A. Sajid¹, J. Steinberg¹ | ¹University of Toronto, ²Michigan State University
Econometric Society World Congress | August 22, 2025

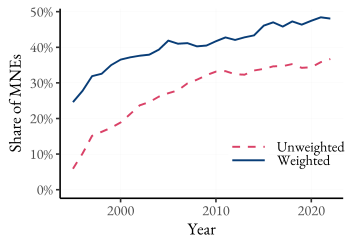
The issue and why do we care?

Introduction

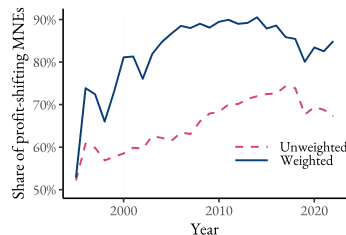
- **Secular Trends**—Since the 1990s, **intangible intensity, multinational production, and profit-shifting** have surged dramatically.



(a) Intangible Intensity



(b) Share of MNE



(c) Share of Profit-shifting MNEs

The issue and why do we care?

Introduction

- **Secular Trends**—Since the 1990s, **intangible intensity, multinational production, and profit-shifting** have surged dramatically.
- **Why It Matters**—most studies analyze each of these trends in isolation:
 - Intangible capital and growth: Crouzet and Eberly (2021), De Ridder (2024)
 - Gains from globalization: Ramondo and Rodríguez-Clare (2013), Burstein and Monge-Naranjo (2024)
 - Tax base erosion through transfer pricing of intangibles: Santacreu (2023), Dyrda, Hong, and Steinberg (2024)

The issue and why do we care?

Introduction

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 - Tax base erosion through transfer pricing of intangibles: Santacreu (2023), Dyrda, Hong, and Steinberg (2024)
- **Our Objective**—unified approach:
Identify the key forces behind the **“intangible revolution”** and its **interaction** with globalization and profit-shifting.

What We Do

Introduction

1. Empirical Linkage

Rising intangible intensity, MNE expansion, and profit-shifting **are linked empirically** using consistent firm-level data. 

2. Structural Linkage

Those same trends **are linked structurally** in a stylized analytical model, where nonrival intangibles, FDI decisions, and profit-shifting interact systemically reinforcing each other.

3. Quantitative Synthesis

The quantitative model integrates these elements, illustrating how **intangible prices and declining FDI barriers** drive the observed co-movements. (in progress)

Empirical Evidence

Cross-sectional relationship

Empirics

Dependent Var	Intangible Intensity					
	(1)	(2)	(3)	(4)	(5)	(6)
MNE indicator	0.09*** (0.01)	0.12*** (0.01)	0.06*** (0.00)			
× post-2005	0.05*** (0.01)	0.05*** (0.01)	0.05*** (0.01)			
TA-region indicator				0.03*** (0.01)	0.04*** (0.01)	0.03*** (0.01)
× post-2005				0.03*** (0.01)	0.04*** (0.01)	0.02** (0.01)
Industry FE	N	N	Y	N	N	Y
Control for log revenue	N	Y	Y	N	Y	Y
Observations	27,318	27,318	27,318	11,925	11,925	11,925

Note: All specifications control for year FE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- MNEs and profit-shifting : ↑ intan. intensity, also true on the intensive margin [▶ intensive](#)
- These gaps have been increasing over time

[▶ details](#)

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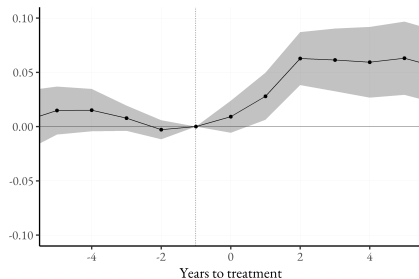
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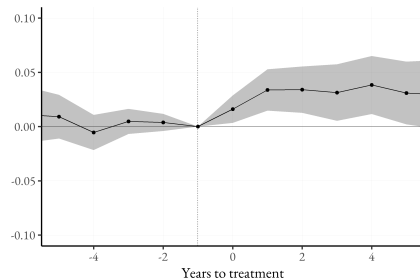
[▶ details](#)

Event studies

Empirics



(a) MNE



(b) Profit Shifting

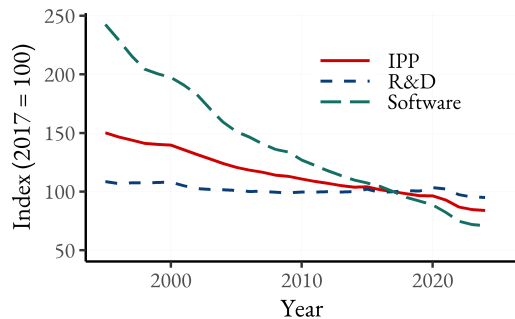
- Expansion into MNE and engagement in profit shifting increases intangible intensity by **≈ 7 and 3 percentage points after 2–4 years**, respectively.

[► details](#)

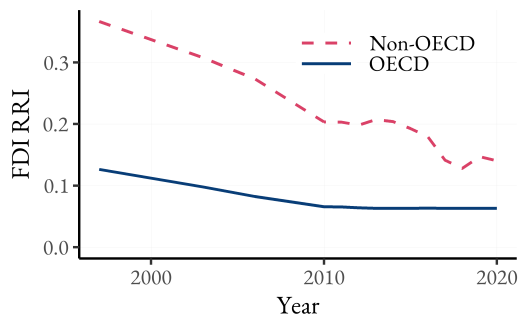
[► add. details](#)

Declining intangibles costs and FDI frictions

Empirics



(a) Intangibles costs



(b) FDI restrictions

Note. BEA Price Index for Private Fixed Investment in Intellectual Property Products (IPP) and OECD FDI Regulatory Restrictiveness Index. Price indices for IPPs are deflated by CPI and normalized to 1 in 2017.

- Both intangibles costs and FDI frictions have been declining over time.

Theory

A theory of intangibles, multinational expansion and profit shifting

Theory

Suppose a world consisting of a set of identical, atomistic productive countries and a tax haven.

Consider the following problem of a firm:

$$\pi(a) = \max_{J, z, \lambda, k} \left\{ (1 - \tau^e(\lambda)) \cdot J\pi(a, l, z) - (1 - \tau) (MC \cdot z + C^\lambda(\lambda, J\pi) + C^J(J)) \right\}$$

s.t.

$$\tau^e(\lambda) = \lambda\tau_{TH} + (1 - \lambda)\tau$$

$$\pi(a, l, z) = pa \left[\mu z^{\frac{\sigma-1}{\sigma}} + (1 - \mu) l^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma\alpha}{\sigma-1}} - wl$$

where

- J is the measure of markets
- z is **nonrival** intangible capital, l is labor
- λ is the share of gross profits shifted to the tax haven
- MC is the marginal cost of z , $C^\lambda(\cdot)$ and $C^J(\cdot)$ are profit shifting and expansion costs
- μ is the weight on the intangibles, σ is the elasticity of substitution, α is the returns-to-scale

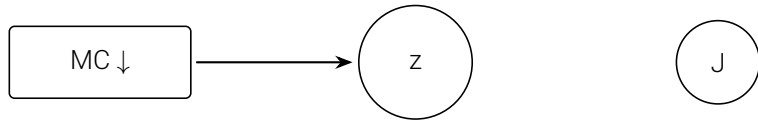
How intangibles z and multinational expansion J reinforce each other

Theory



How intangibles z and multinational expansion J reinforce each other

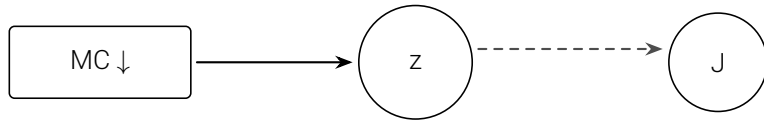
Theory



- Firms **substitute** towards intangible capital ($z \uparrow$) as the MC of intangibles decreases ($MC \downarrow$)

How intangibles z and multinational expansion J reinforce each other

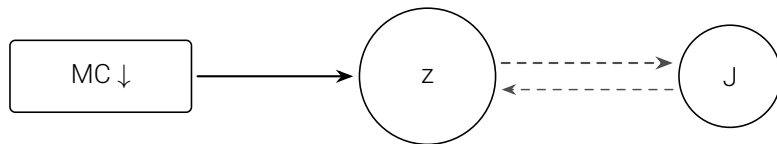
Theory



- Firms **substitute** towards intangible capital ($z \uparrow$) as the MC of intangibles decreases ($MC \downarrow$)
- The **non-rivalry** of intangible capital incentivizes multinational expansion ($z \uparrow \implies J \uparrow$).

How intangibles z and multinational expansion J reinforce each other

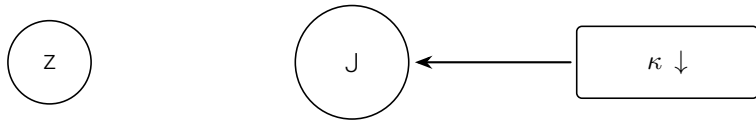
Theory



- Firms **substitute** towards intangible capital ($z \uparrow$) as the MC of intangibles decreases ($MC \downarrow$)
- The **non-rivalry** of intangible capital incentivizes multinational expansion ($z \uparrow \implies J \uparrow$).
- Greater multinational reach ($J \uparrow$) increases the return to intangible capital, **reinforcing** the impact on intangibles ($z \uparrow$).

How intangibles z and multinational expansion J reinforce each other

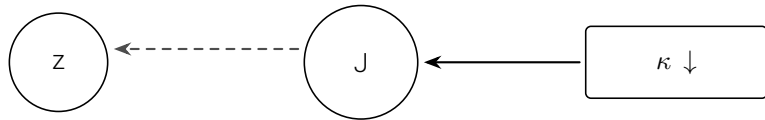
Theory



- Falling expansion costs ($\kappa \downarrow$) increases firm incentives to **expand** ($J \uparrow$)

How intangibles z and multinational expansion J reinforce each other

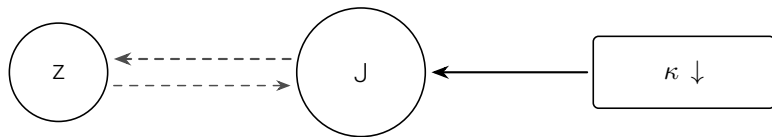
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How intangibles z and multinational expansion J reinforce each other

Theory



- Falling expansion costs ($\kappa \downarrow$) increases firm incentives to **expand** ($J \uparrow$)
- Greater multinational reach increases the **return to intangible capital** ($J \uparrow \implies z \uparrow$)
- A tilt towards intangibles ($z \uparrow$) makes MNE expansion even **more attractive** ($J \uparrow$)

How intangibles z and multinational expansion J reinforce each other

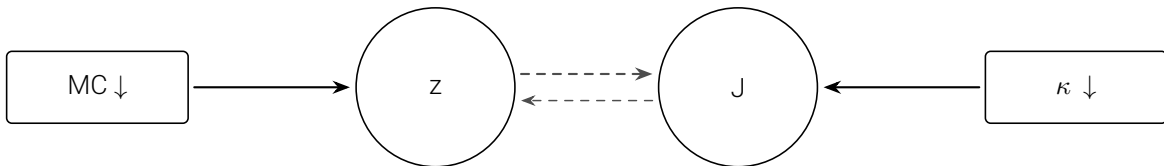
Theory



- **Concurrently** falling MC of intangible capital and expansion costs have the direct effects of tilting firms towards intangible capital and increasing multinational expansion

How intangibles z and multinational expansion J reinforce each other

Theory



- **Concurrently** falling MC of intangible capital and expansion costs have the direct effects of tilting firms towards intangible capital and increasing multinational expansion
- Mutual reinforcement results in greater z and J than each decrease in isolation – **complementarity**.

Summary of theoretical results

Theory

- **Monotonicity:**

1. Higher productivity a firms have higher z and J .
2. Decreases in MC , κ , and ι all increase z and J .

- **Complementarity:**

1. The effects of decreases in MC , κ , and ι on z and J are greater for high productivity firms.
2. The effects of decreases in MC , κ , and ι on z and J are complementarity to each other.

Quantitative Model

Dynamic, heterogeneous firms model

Quantitative Model

Core features:

- A firm is a triplet of individual states: (a, z, J)
 - a : stochastic, persistent productivity
 - z : intangible capital, nonrival within a firm
 - J : number of foreign subsidiaries
- Choices:
 - Intangible investment subject **marginal cost** MC and quadratic adjustment costs.
 - Rival production factors (capital, labor).
 - Whether to expand or contract across locations subject to **costs** κ . Expansion affected by **FDI barriers** ψ .
- Aggregation:
 - Distribution of firms over individual states.
 - Endogenous selection splits firm into two sets: non-MNES ($J = 0$) and MNEs ($J > 0$).

Model discipline and key objects of interests

Quantitative Model

Discipline for key parameters:

- κ disciplined by fraction of MNEs: 40% in our data set.
- μ disciplined by the average intangible intensity: 48% in 2022.
- σ (elasticity of substitution in production) identified by cross-sectional differences in intangible intensity between MNEs and non-MNEs.

Exercises in parametrized economy:

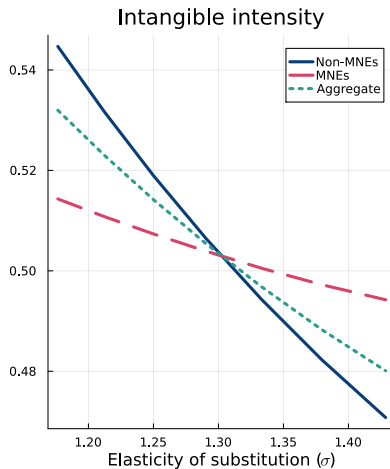
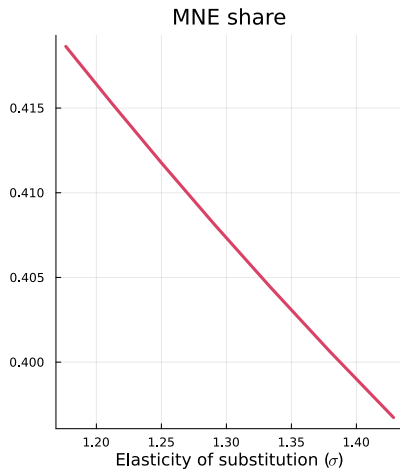
- Reduction in marginal cost of intangible investment mimicking declining R&D.
- Decline in FDI restrictions reflecting drop in FDI barriers over time.
- Both decline at the same time to inspect the degree of complementarity.

Objects of interest:

- Share of MNEs
- Intangible intensity: aggregate, for MNEs and for non-MNEs

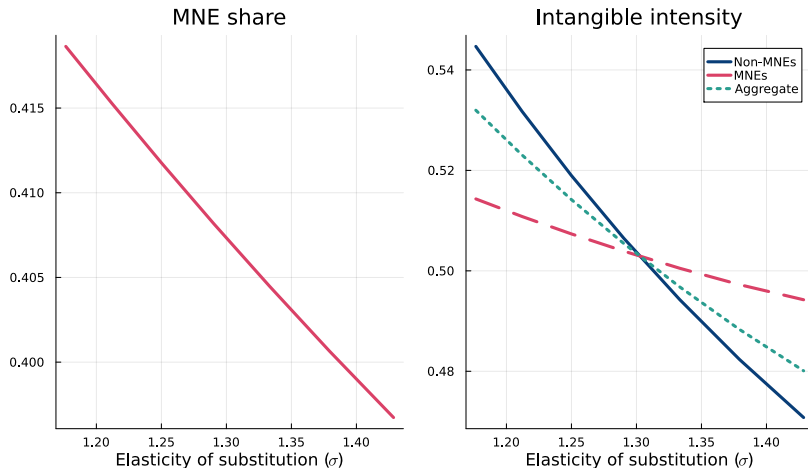
Identification of the elasticity of substitution σ

Quantitative Model



Identification of the elasticity of substitution σ

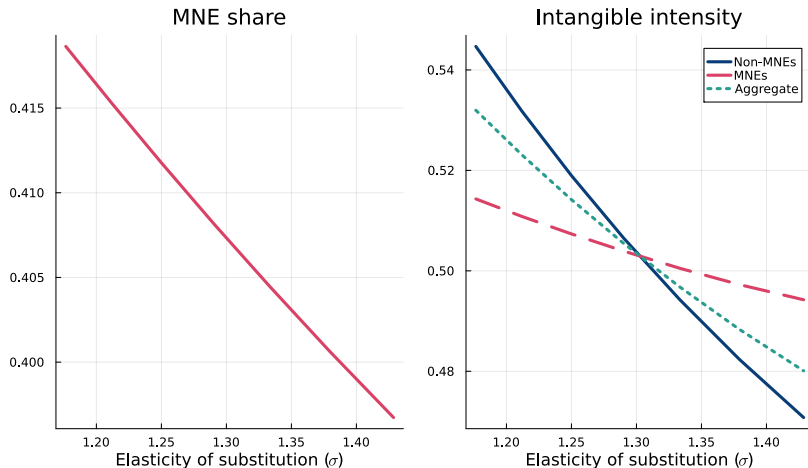
Quantitative Model



Elasticity of intangible share with respect number of affiliates is increasing in σ

Identification of the elasticity of substitution σ

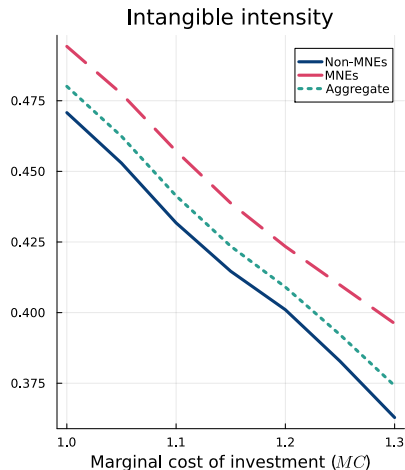
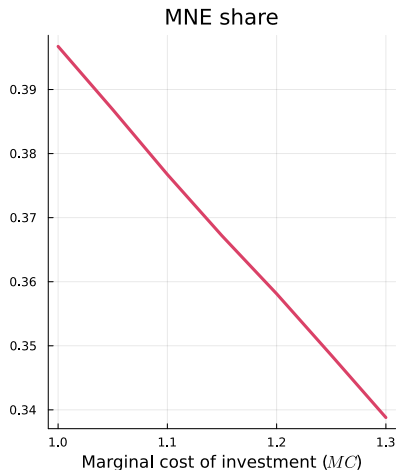
Quantitative Model



Allows us to identify σ from cross-sectional comparison of MNEs vs non-MNEs

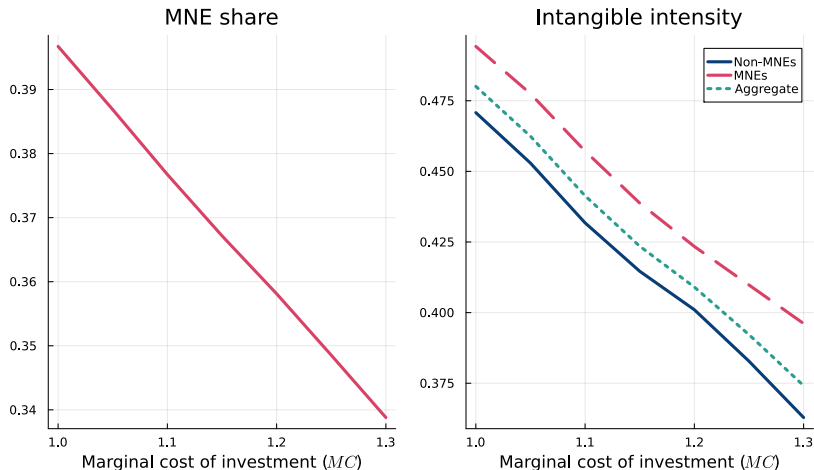
Fall in the marginal cost of intangible investment MC

Quantitative Model



Fall in the marginal cost of intangible investment MC

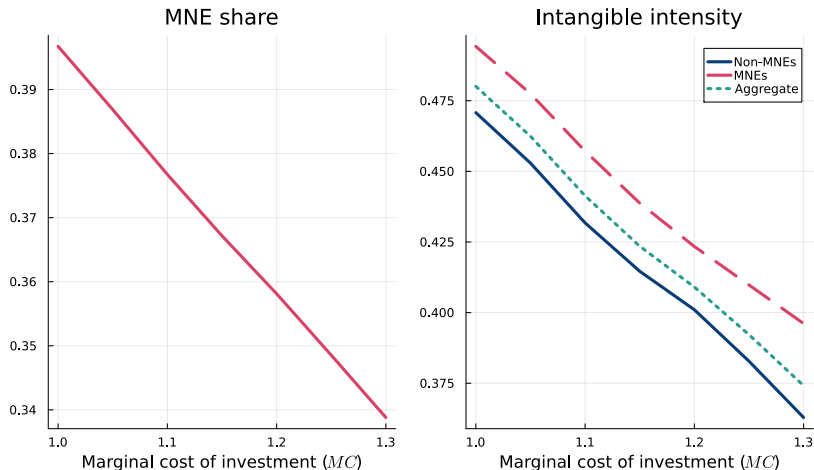
Quantitative Model



Intensive margin: intangible intensity falls for both MNEs and non-MNEs

Fall in the marginal cost of intangible investment MC

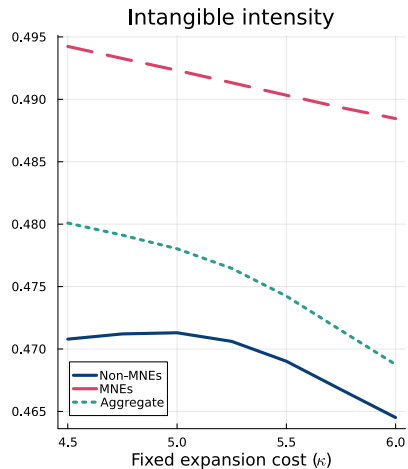
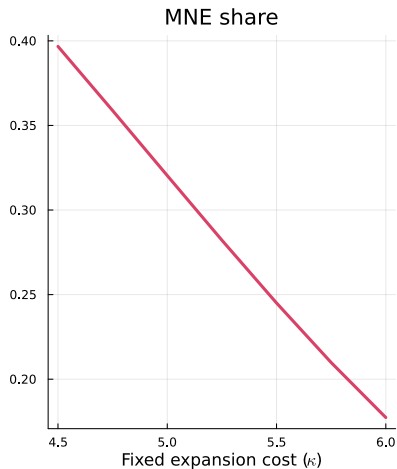
Quantitative Model



Ext. margin: MNE share falls. MNEs have greater intensity, so agg. intensity falls (slightly) faster.

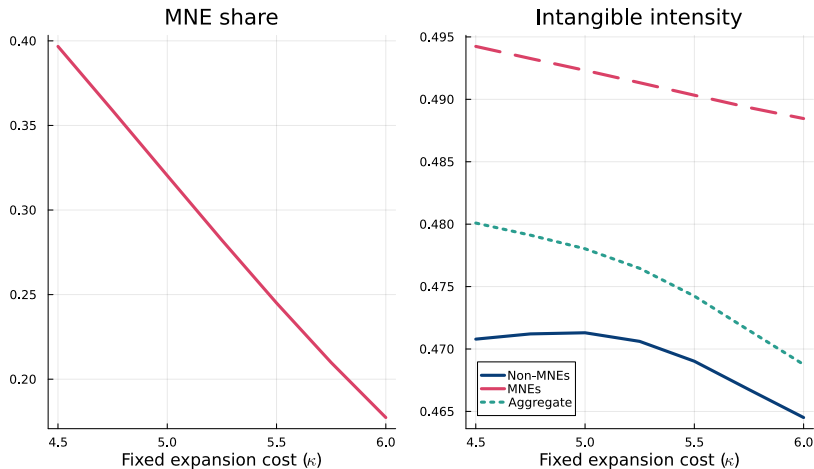
Fall in the expansion costs κ

Quantitative Model



Fall in the expansion costs κ

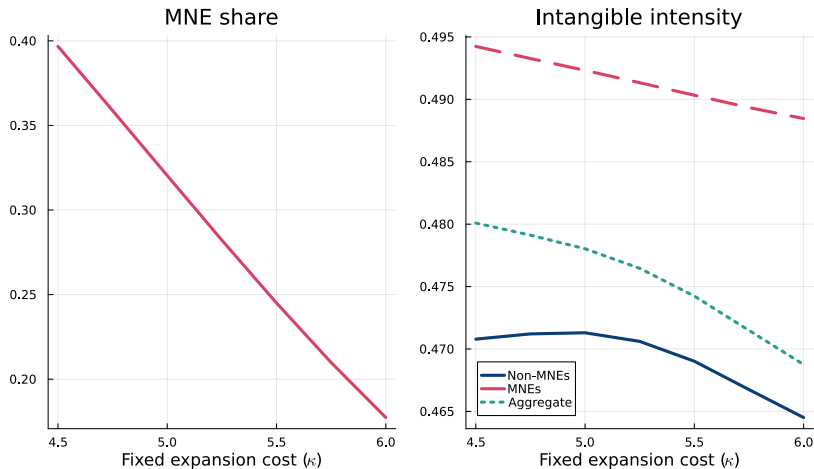
Quantitative Model



Operates primarily through the extensive margin/composition effect.

Fall in the expansion costs κ

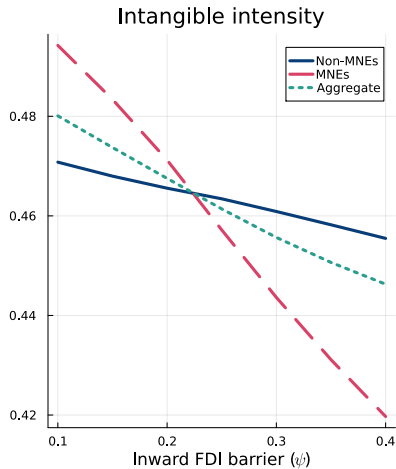
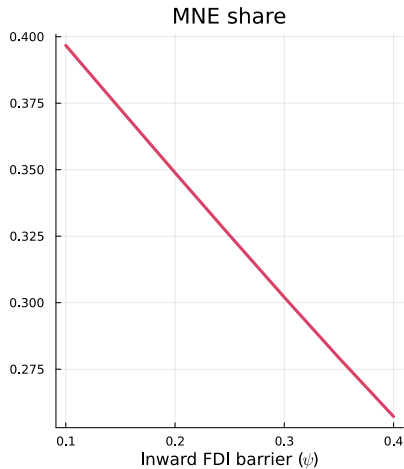
Quantitative Model



Also affects non-MNEs at the intensive margin—higher odds of becoming MNE in the future.

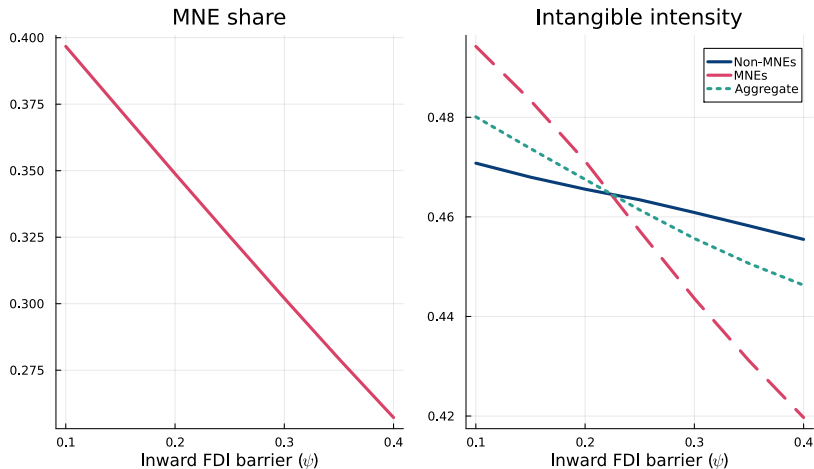
Fall in the FDI barriers ψ

Quantitative Model



Fall in the FDI barriers ψ

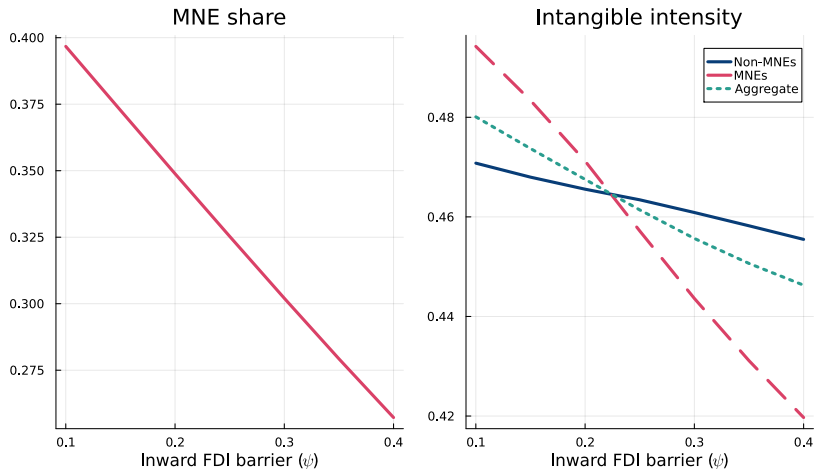
Quantitative Model



The impact on intensity much larger for MNEs.

Fall in the FDI barriers ψ

Quantitative Model



Non-MNEs: (1) lower odds of becoming MNE (2) less need to scale up

Testing complementarities: MC and κ

Quantitative Model

	High Investment Cost	Low Investment Cost
	% changes from High-High	
Panel (A): MNE share		
High Expansion Cost	–	0.34
Low Expansion Cost	1.56	2.12
Panel (B): Intangible Intensity		
High Expansion Cost	–	0.32
Low Expansion Cost	0.05	0.46

Testing complementarities: MC and κ

Quantitative Model

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Fall in expansion cost and drop in MC in drive up MNE share and intangible intensity, and...

Testing complementarities: MC and κ

Quantitative Model

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... their joint effect is larger than the sum of the two (complementarity).

Key Takeaways

What to Remember

1. **Three defining trends** since the 1990s:
 - Intangible intensity ↑
 - Multinational reach ↑
 - Profit-shifting ↑
2. **Empirical link:** Cross-sectional regressions and firm-level event studies confirm these moves are tightly correlated.
3. **Mechanism:** Falling R&D costs and lower FDI barriers jointly drive more intangibles and more foreign affiliates.
4. **Complementarity:** These forces reinforce each other—the combined effect exceeds the sum of their individual impacts.

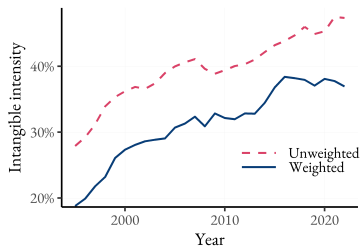
Next step: use the model to run a horse race between the exogenous driving forces to decompose their effects on the endogenous trends of interest.

Appendix

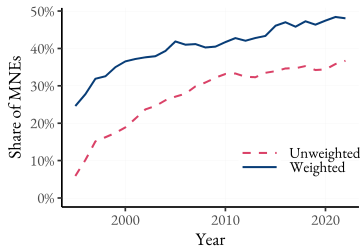
- Firm balance-sheet data from Compustat North America
 - Consolidated financial statement
 - Information on sales, tangible and intangible capital, investment, foreign income, and tax liabilities
- Intangible capital
 - Measured using the perpetual inventory method (Peters and Taylor, 2017)
- Multinational activity
 - Extensive margin: firms that have foreign subsidiaries reported in Exhibit 21 of 10-K forms
 - Intensive margin: the number of foreign countries and the share of foreign income
- Profit-shifting measures
 - Extensive margin: firms with subsidiaries in the tax-haven, identified by Tørsløv et al., (2022)
 - Intensive margin: share of profits shifted estimated by Delis et al. (2025)

Rise in intangibles, MNE, and profit shifting [▶ return](#)

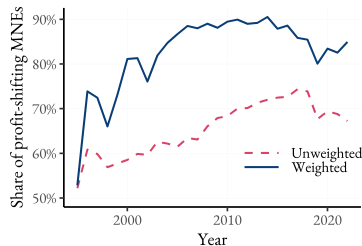
Empirics



(a) Intangible Intensity



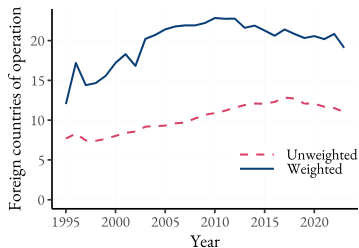
(b) Share of MNE



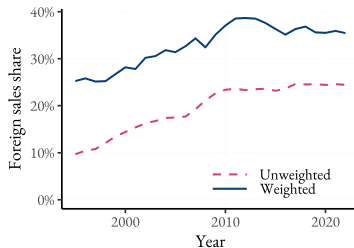
(c) Share of Profit-shifting MNEs

- Intangible Intensity = $\text{Intangible capital} / (\text{Intangible capital} + \text{Tangible capital})$
 - figure (a) is robust to (1) using intangible-revenue ratio, (2) using only knowledge capital, (3) controlling for industry composition [▶ industry](#)
- Intensive margin: ↑ number of foreign subsidiaries and profit-shifting intensity [▶ intensive](#)

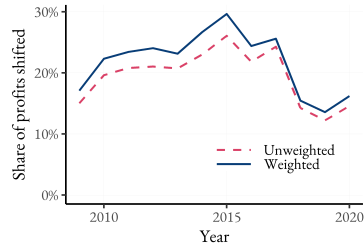
Intensive margin: multinational expansion and profit shifting [▶ return](#)



(a) Avg # Foreign Subsidiaries



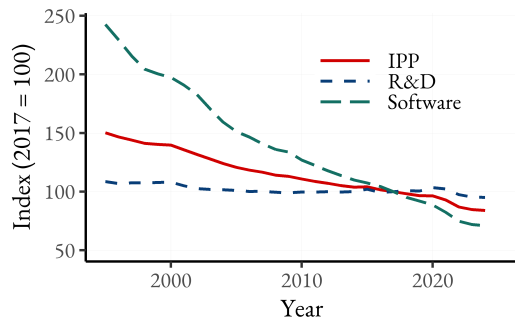
(b) Share of Foreign Income



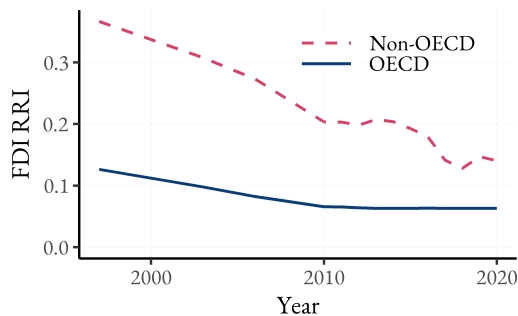
(c) Share of Shifted Profits by Delis et al. (2025)

Declining intangibles costs and FDI frictions

Empirics



(a) Intangibles costs



(b) FDI restrictions

Note. BEA Price Index for Private Fixed Investment in Intellectual Property Products (IPP) and OECD FDI Regulatory Restrictiveness Index. Price indices for IPPs are deflated by CPI and normalized to 1 in 2017.

- Both intangibles costs and FDI frictions have been declining over time.

Details on the cross-sectional regressions [▶ return](#)

We estimate the following regression

$$\text{Intangible intensity}_{jt} = \alpha + \beta_1 1_{\{Post-2005_{jt}\}} + \beta_2 1_{\{X_{jt}\}} + \beta_3 1_{\{Post-2005_{jt}\}} \times 1_{\{X_{jt}\}} + Z_{jt}\delta + \varepsilon_{jt}$$

- $1_{\{X_{jt}\}}$ is an indicator for whether firm j in year t is an MNE or has a subsidiary in a tax-avoidance region
- $1_{\{Post-2005_{jt}\}}$ is a post-2005 time indicator. More flexible time trend indicators yield similar results.
- Z_{jt} is a vector of controls. In the fullest specifications, we include:
 - year FEs to account for the secular increase in intangible intensity over our analysis period,
 - industry FEs to account for systematic variation in intangible intensity across industries (for e.g., differences in production technology, or systematic over/underreporting of intangibles on firm balance sheets from accounting conventions), &
 - log firm revenue to proxy for time-varying firm characteristics, such as idiosyncratic productivityyear fixed-effects, industry fixed-effects, and log firm revenue.
- Estimates for β_2 for each X_{jt} are reported, as well as the interaction with the post-2005 indicator, β_3

Details on the event study [▶ return](#)

We run the following event-study regression

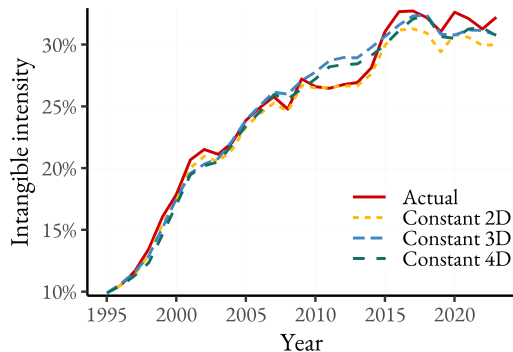
$$\text{Intangible intensity}_{jt} = \gamma_j + \delta_t + \sum_{e \notin C} \sum_{\tau \neq -1} \beta_{e,\tau} \cdot 1_{\{E_j=e\}} \cdot 1_{\{t-E_i=\tau\}} + \varepsilon_{jt},$$

- E_j is the year when firm j becomes an MNE or opens the first tax-haven subsidiary
- $\beta_{e,\tau}$ is the cohort-specific average treatment effect on the treated (CATT) τ periods from initial treatment for the cohort of firms first treated in year e
 - The dynamic treatment effect estimate for each relative time period is given by the average of $\hat{\beta}_{e,\tau}$, weighted by the proportion of each cohort e in the relative time period τ
- γ_j and δ_t are firm and year fixed effects

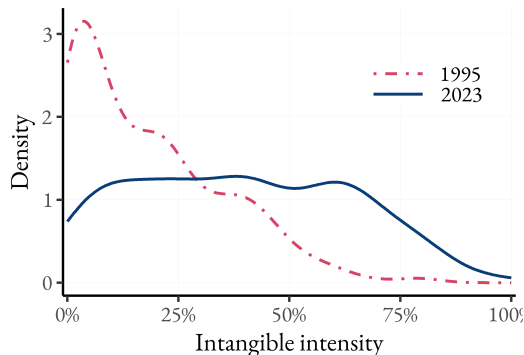
Details on the event study [▶ return](#)

- The “building blocks” for the event study estimates are the cohort-specific average treatment effect on the treated, $\beta_{e,t}$, which are identified under a parallel trends and no anticipation assumption.
- $\beta_{e,t}$ is then averaged across all cohorts e for a given relative time, t , using weights that are the sample share of each cohort e
- The IW estimator is robust to treatment effect heterogeneity, which is potentially problematic for standard TWFE regressions.
- The parallel trends assumption seems feasible in this context since the leads of the treatment are statistically insignificant from zero.

More on Intangible Intensity with [return](#)



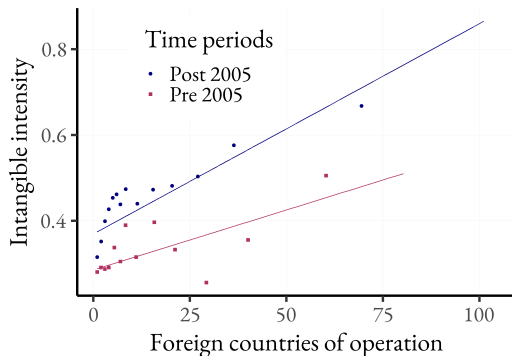
(a) Intang. intensity w. fixed ind. composition



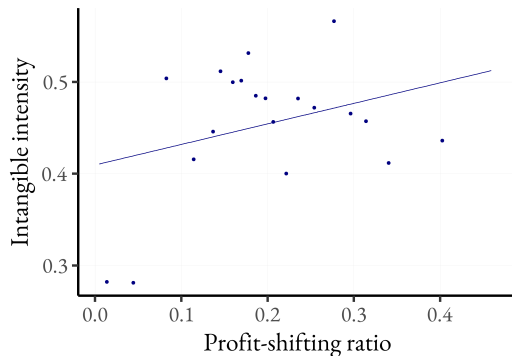
(b) Density plot

- Intangible intensity rises mainly within industries, rather than shifts towards high-intangible industries

Intensity margin: intangible intensity vs expansion and profit shifting



(a) Number of foreign countries



(b) Share of Shifted Profits by Delis et al. (2025)

- The firm-year level shares of shifted profits are only estimated since 2009

Cross-sectional relationship

Empirics

Dependent Var	Intangible Intensity					
	(1)	(2)	(3)	(4)	(5)	(6)
MNE indicator	0.08*** (0.01)	0.04*** (0.01)	0.08*** (0.01)			
× post-2005	0.06*** (0.01)	0.04*** (0.01)	0.03*** (0.01)			
TA-region indicator				0.06*** (0.01)	0.03*** (0.01)	0.07*** (0.01)
× post-2005				0.05*** (0.01)	0.03*** (0.01)	0.02*** (0.01)
Industry FE	N	Y	Y	N	Y	Y
Control for log revenue	N	N	Y	N	N	Y

Note: All specifications control for year FE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- MNEs and profit-shifting : $\uparrow$ intan. intensity, also true on the intensive margin [▶ intensive](#)
- These gaps have been increasing over time

Cross-sectional relationship

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Intuition behind monotonicity and complementarity

Theory

- From the first-order condition of z and l :

$$s_z \equiv \frac{MC \cdot z}{Jwl} = \left(\frac{1-\mu}{\mu} \tilde{\tau}(\lambda) \right)^{-\sigma} \left(\frac{w}{MC} \right)^{\sigma-1} J(MC, \kappa, \iota; a)^{\sigma-1}$$

when $\sigma > 1$, i.e. z and l are gross substitutes:

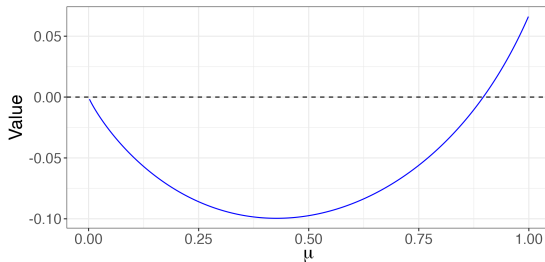
- Standard result: **plant-level substitution** towards z when $MC \downarrow$
 - With non-rivalry: **firm-level substitution** towards z as $J \uparrow$ (when $\kappa \downarrow$ and $a \uparrow$)
 - With profit-shifting: increases the **after-tax return** of z through $\downarrow$ in $\tilde{\tau}(\lambda)$ (when $\iota \downarrow$)
- Complementarity between MC , ι and κ :
 - e.g. the effect of $\downarrow MC$ on s_z depends on its impact on J
 - thus, the total effect of MC on s^z is greater when κ is smaller, as $\frac{\partial^2 \ln J}{\partial \ln \kappa \partial \ln MC} > 0$

Take the cross partial derivatives of κ and MC as an example. We have

$$\frac{\partial^2 \ln J}{\partial \ln \kappa \partial \ln MC} = (-1) \frac{1}{\left(\gamma - 1 - \frac{\partial \ln \pi}{\partial \ln J}\right)^2} \cdot \frac{\partial^2 \ln \pi}{\partial \ln J \partial \ln MC}$$

$$\frac{\partial^2 \log s^z}{\partial \ln \kappa \partial \ln MC} = (\sigma - 1) \frac{\partial^2 \ln J}{\partial \ln \kappa \partial \ln MC}$$

Plot $\frac{\partial^2 \ln \pi}{\partial \ln J \partial \ln MC}$ against μ :



Assumption and definition

Theory

Assumption

1. The expansion cost function is $C^J(J) = \kappa \cdot J^\gamma$ with $\gamma - 1 > \frac{\partial \ln \pi}{\partial \ln J}$, i.e. the marginal cost of expansion increases faster than the gross profits per market as J increases.
2. The profit shifting costs is $C^\lambda(\lambda, J\pi) = \frac{1}{2}\iota\lambda^2 \cdot J\pi$.

Definition

The **combined elasticity** of x with respect to x_1 and x_2 is defined as

$$\varepsilon_{y,(x_1,x_2)} = \frac{\partial \ln y}{\partial \ln x_1} + \frac{\partial \ln y}{\partial \ln x_2} + \frac{\partial^2 \ln y}{\partial \ln x_1 \partial \ln x_2}$$

Intangible intensity and multinational expansion

Theory

Proposition (Monotonicity)

Let $s^z \equiv \frac{MC \cdot z}{J_{wl}}$ be the intangible intensity. The following statements are true:

1. A decrease in MC , in ι , or in κ **increases** the optimal market choice J :

$$\varepsilon_{J,x} < 0,$$

where $x \in \{MC, \iota, \kappa\}$.

2. When $\sigma > 1$, a decrease in MC , in ι , or in κ **increases** the intangible intensity s^z :

$$\varepsilon_{s^z,x} < 0,$$

where $x \in \{MC, \iota, \kappa\}$.

Selection, intangible intensity and multinational expansion

Theory

Proposition (Monotonicity in firm productivity)

Let $s^z \equiv \frac{MC \cdot z}{J_{wl}}$ be the intangible intensity. The following statements are true:

1. Higher productivity firms operate in more foreign markets J :

$$\varepsilon_{J,a} > 0,$$

2. When $\sigma > 1$, higher productivity firms have greater intangible intensity s^z :

$$\varepsilon_{s^z,a} > 0.$$

Complementarity

Theory

Proposition (Complementarity)

The following statements are true except when $\mu \rightarrow 1$: [▶ detail](#)

1. The positive effects of reductions in MC , ι , and κ on J are **complementary** to each other:

$$\varepsilon_{J,(x_1,x_2)} > \varepsilon_{J,x_1} + \varepsilon_{J,x_2}$$

where $x_1, x_2 \in \{MC, \iota, \kappa\}$.

2. When $\sigma > 1$, the positive effects of reductions MC , ι , and κ on s^z are **complementary** to each other:

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Complementarity with firm productivity

Theory

Proposition (Complementarity with firm productivity)

The following statements are true except when $\mu \rightarrow 1$: [detail](#)

1. The positive effects of reductions in MC , ι , and κ on J are **greater** for high-productivity firms:

$$\varepsilon_{J,(a,x)} < \varepsilon_{J,a} + \varepsilon_{J,x}$$

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where $x \in \{MC, \iota, \kappa\}$.

Testing complementarities: MC and κ

Quantitative Model

	High Investment Cost	Low Investment Cost
	% changes from High-High	
Panel (A): MNE share		
High Expansion Cost	–	0.34
Low Expansion Cost	1.56	2.12
Panel (B): Intangible Intensity		
High Expansion Cost	–	0.32
Low Expansion Cost	0.05	0.46

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Similarly fall in expansion cost and drop in MC in drive up MNE share and intangible intensity, and...

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... their joint effect is larger than the sum of the two (complementarity).

Testing complementarities: MC and ψ

Quantitative Model

	High Investment Cost	Low Investment Cost
	% changes from High-High	
<i>Panel (A): MNE share</i>		
High FDI Barriers	–	0.17
Low FDI Barriers	0.54	0.84
<i>Panel (B): Intangible Intensity</i>		
High FDI Barriers	–	0.29
Low FDI Barriers	0.08	0.45

Testing complementarities: MC and ψ

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Both fall in FDI barriers and drop in MC in drive up MNE share and intangible intensity, but...

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